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Savitribha Institute of Medical And Technical Science
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Smart Water Quality Monitoring and Distribution Management System

DSA0104-Object Oriented Programming with C++

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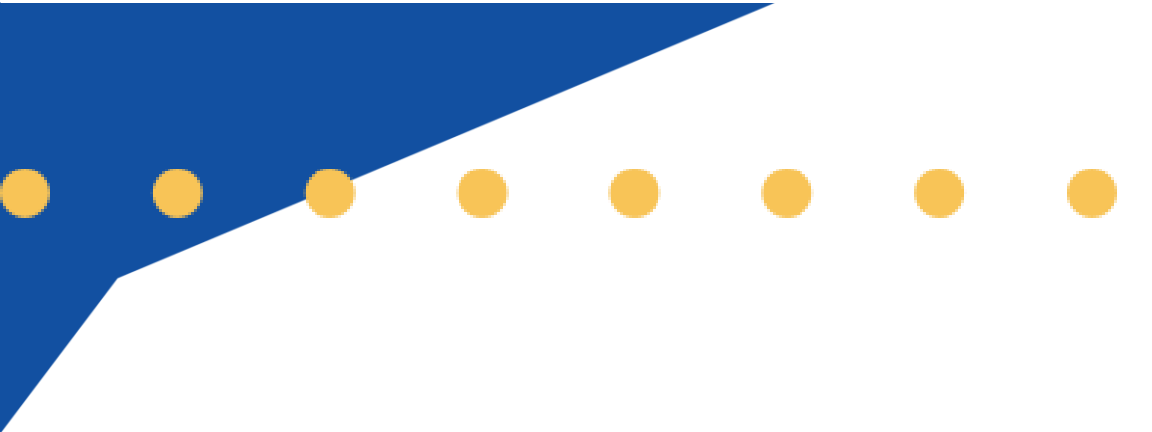
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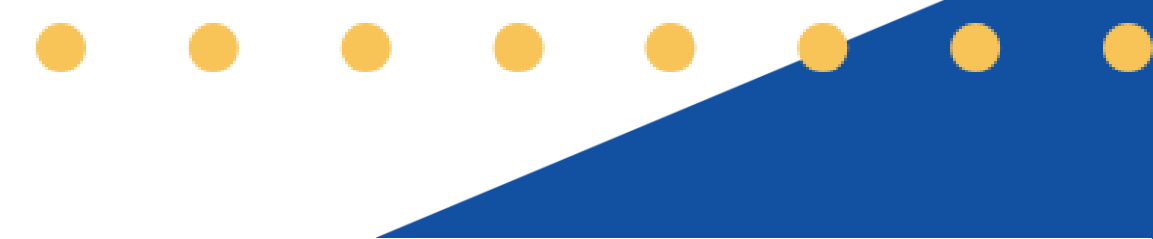
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Introduction

- Smart Water Quality Monitoring and Distribution Management System ensures safe water supply.
- The system monitors pH, temperature, turbidity, and TDS values continuously.
- It tracks water levels in tanks and storage reservoirs efficiently.
- Water distribution is managed to minimize wastage and optimize supply.
- Real-time monitoring improves water quality and operational performance.
- The system stores data and generates reports for analysis.
- It supports sustainable water management and smart city development.
- Automated monitoring helps detect issues and maintain reliable distribution.



Objective

- Monitor water quality parameters continuously to ensure safe drinking water supply.
 - Detect contamination and generate alerts for immediate corrective actions.
 - Monitor tank water levels and prevent overflow or shortages.
 - Optimize water distribution based on demand and resource availability.
 - Reduce water wastage and improve overall management efficiency.
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Problem Statement

- Manual monitoring can delay detection of water contamination and quality issues.
- Poor monitoring of water levels and distribution can cause shortages, overflow, leakage, and wastage.
- A real-time smart system is needed to monitor water quality, storage, and distribution efficiently.
- The system should provide automatic alerts and optimized water management for a reliable supply.

Literature Survey

LITERATURE REVIEW OF IOT-BASED WATER QUALITY & DISTRIBUTION SYSTEMS

S.No.	Author & Year	Title / System Studied	Techniques / Components Used	Findings / Results	Limitations / Research Gap
1	M. M. S. Islam et al., 2020	Smart Water Quality Monitoring System with Cost-Effective IoT	pH sensor, turbidity sensor, temperature sensor, water-level sensor, MCU, ThingSpeak	Developed an IoT-based system for continuous monitoring of water quality and tank conditions with cloud-based data visualization.	Mainly focused on monitoring; advanced distribution control and leakage management were not fully integrated.
2	Savel Va, 2022	IoT-Based Water Quality Monitoring System Using LoRaWAN	Arduino, LoRaWAN, pH, turbidity, temperature, dissolved oxygen, electrical conductivity, GPS, cloud platform	Enabled long-range collection and cloud monitoring of multiple water-quality parameters using LoRaWAN.	Communication and sensor infrastructure can increase system complexity; distribution optimization was not the main focus.
3	Dhruba et al., 2023	IoT-Based Water Quality Assessment System for Industrial Waste Water	IoT sensors, pH, TDS, turbidity, temperature, mobile application	Demonstrated real-time monitoring of important water-quality parameters and mobile-based access to monitoring information.	Focused mainly on water-quality assessment and did not provide complete storage and distribution management.
4	Pagano et al., 2024	SWI-FEED: Smart Water IoT Framework for Water Distribution Systems	IoT, LoRaWAN, EPANET, NS-3, wireless sensors, simulation framework	Studied large-scale IoT deployment for water-distribution monitoring, including leakage detection, energy consumption, and network performance.	Framework is mainly focused on large-scale network evaluation; integrated water-quality monitoring is not its primary objective.
5	Verma, Nair & Choudhary, 2025	IoT-Enabled Smart Water Distribution System for Urban Areas	Flow sensors, pressure sensors, IoT gateways, cloud platform, data analytics	Integrated real-time monitoring, automated control, leak detection, predictive maintenance, and demand-based water distribution.	Requires reliable sensor connectivity and cloud infrastructure; a unified system combining quality, storage, and distribution can be further developed.

Module 1

Water Quality Monitoring

- **Description**

- ☐ This module focuses on monitoring water quality parameters such as pH, temperature, turbidity, and dissolved solids to ensure safe and clean water distribution. It continuously analyzes sensor data, detects contamination, and generates alerts whenever abnormal water conditions are identified.

- **Activities**

- ☐ Collect pH, temperature, turbidity, and TDS values from sensors.
- ☐ Compare sensor readings with predefined water quality standards.
- ☐ Detect contamination and generate warning notifications automatically.
- ☐ Store monitoring data for analysis and future reporting.
- ☐ Ensure safe water quality before distribution to consumers.

- **Output**

- ☐ Real-time water quality reports with contamination alerts and safety status.

Module 2

Water Storage and Monitoring

- **Description**

- ☐ This module focuses on monitoring water levels in storage tanks and reservoirs to prevent overflow and shortages. It continuously tracks water availability, generates low-level alerts, and maintains records to ensure efficient storage management and uninterrupted water supply.

- **Activities**

- ☐ Monitor water levels in storage tanks and reservoirs continuously.
- ☐ Display real-time information about available water quantity.
- ☐ Detect low-water conditions and generate alert notifications.
- ☐ Prevent tank overflow and water shortages automatically.
- ☐ Store water-level records for future analysis and planning.

- **Output**

- ☐ Real-time water-level reports with storage status and warning alerts.

Module 3

Water Distribution Management

- **Description**

- This module focuses on distributing water efficiently to residential, industrial, and agricultural areas based on demand. It controls water flow through pipelines, detects leakage, and ensures uninterrupted supply by optimizing distribution operations.

- **Activities**

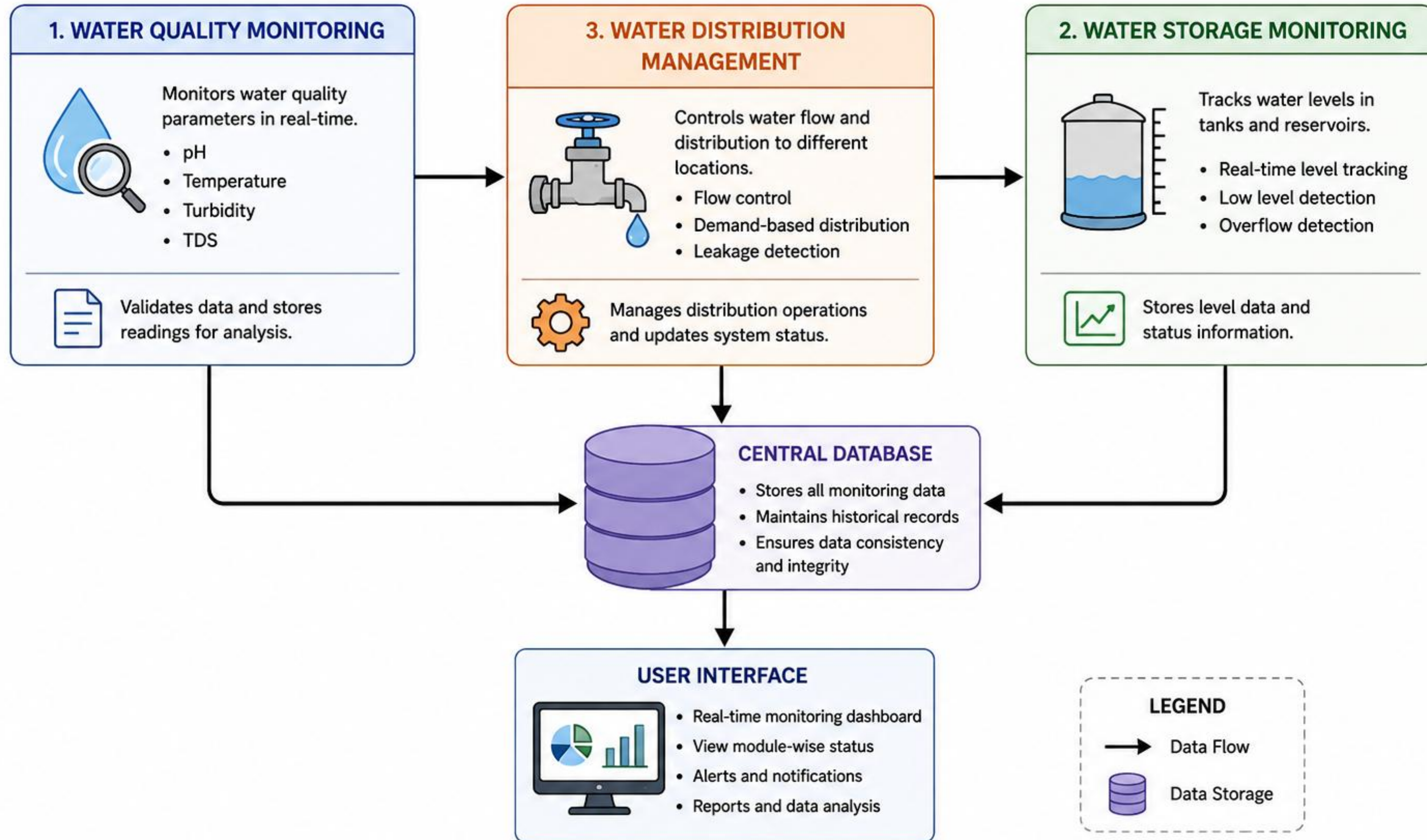
- ☐ Control water flow to different locations using smart valves.
- ☐ Monitor pipeline conditions and detect water leakages automatically.
- ☐ Prioritize water supply based on demand and requirements.
- ☐ Optimize water distribution to reduce wastage and losses.
- ☐ Generate reports on water consumption and distribution patterns.

- **Output**

- ☐ Real-time distribution reports with leakage alerts and supply status.

SYSTEM ARCHITECTURE DIAGRAM

Smart Water Quality Monitoring and Distribution Management System (Software Project)



Conclusion

- The system continuously monitors water quality parameters to ensure safe water supply.
- It efficiently tracks water levels in tanks and reservoirs.
- The distribution module optimizes water flow and minimizes wastage.
- Real-time monitoring improves resource management and operational efficiency.
- The project supports sustainable and intelligent water distribution systems.
- It provides accurate reports for better decision-making and planning.
- The system helps prevent contamination and ensures reliable water management.
- Smart monitoring techniques improve overall water quality and distribution performance.



**Thank
You**